



COMP90050: Advanced Database Systems

Lecturer: Farhana Choudhury (PhD)
Different cloud services (from Amazon)
Week 1



Amazon Elastic Compute Cloud (EC2)

📦 Amazon EC2 = Virtual Machine

📦 Amazon EC2: on-demand compute power

- Obtain and boot new server instances in minutes
- Quickly scale capacity up or down
- Servers from \$0.02 (2 cents) per hour
- On Demand, Reserved, and Spot Pricing

📦 Key features:

- Support for Windows, Linux, FreeBSD, and OpenSolaris
- Supports all major web and application platforms
- Deploy across Availability Zones for reliability
- monitors status and usage

Amazon Elastic Block Store (EBS)

- 📦 You can use Amazon EBS as you would use a hard drive on a physical server.
- 📦 Amazon EBS is particularly well-suited for use as the primary storage for a file system, database or for any applications that require fine granular updates and access to raw, unformatted block-level storage.

Amazon Simple Storage Service (S3)

- ❏ In traditional on-premise applications, this type of data would ordinarily be maintained on **SAN** or **NAS**. However, a cloud-based mechanism such as Amazon S3 is far more agile, flexible, and geo-redundant.
- ❏ Amazon S3 is a highly scalable, durable and available distributed object store designed for mission-critical and primary data storage with an easy to use web service interface.



Amazon S3

Amazon EC2

Amazon VPC

Amazon Elastic
MapReduceAmazon
CloudFront

Amazon RDS

Amazon SNS

Navigation

Region: US-E

> VPC Dashboard

VIRTUAL PRIVATE

> Your VPC

> Subnets

> Routing

> Internet Gate

> DHCP Option

> Elastic IPs

SECURITY

> Network ACLs

> Security Groups

VPN CONNECTION

> Customer Gate

> VPN Gateway

> VPN Connect

Create a Virtual Private Cloud

Cancel

Select a VPC Quickstart configuration below:

☐ VPC with a Single Public Subnet

Your instances run in a private, isolated section of the AWS cloud with direct access to the Internet. Network access control lists and security groups can be used to provide strict control over inbound and outbound network traffic to your instances.

☐ VPC with Public & Private Subnets

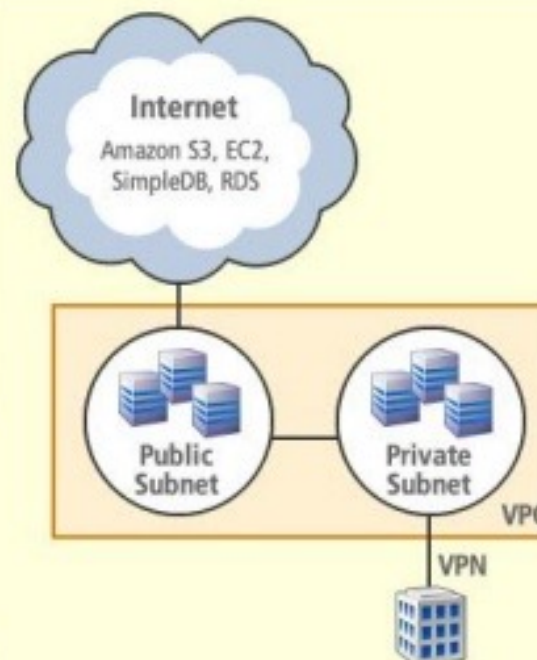
This configuration adds a second subnet whose instances are not exposed to the Internet. Instances in this subnet communicate with the Internet via Network Address Translation.

☒ VPC with Internet & VPN Access

This configuration adds an IPsec Virtual Private Network (VPN) connection between your VPC and your data center - extending your data center to the cloud while also providing direct access to the Internet for instances in your VPC. [View details](#)

☐ VPC with VPN Only Access

Your instances run in a private subnet which is connected to your corporate data center via an IPsec VPN connection. All communication with the Internet is routed via the VPN connection and out your data center. This configuration has no direct access to the Internet.



Creates: a /16 network with two /24 subnets. One subnet is directly connected to the Internet while the other subnet is connected to your corporate network via IPsec VPN tunnel. (VPN charges apply)

Continue